Ionic bonding

Introduction

Ionic bond: the strong electrostatic attraction between oppositely charged ions formed when a metal reacts with a non-metal.

In an ionic compound the ionsd are held in a repeating pattern (alternating +ive and −ive) called a giant ionic lattice structure where the attraction force between oppistve charges is maximized while repulsion between similar ones is reduced.

note: ion attration forces do not rely on oriantation nor dircection (omidirectional)

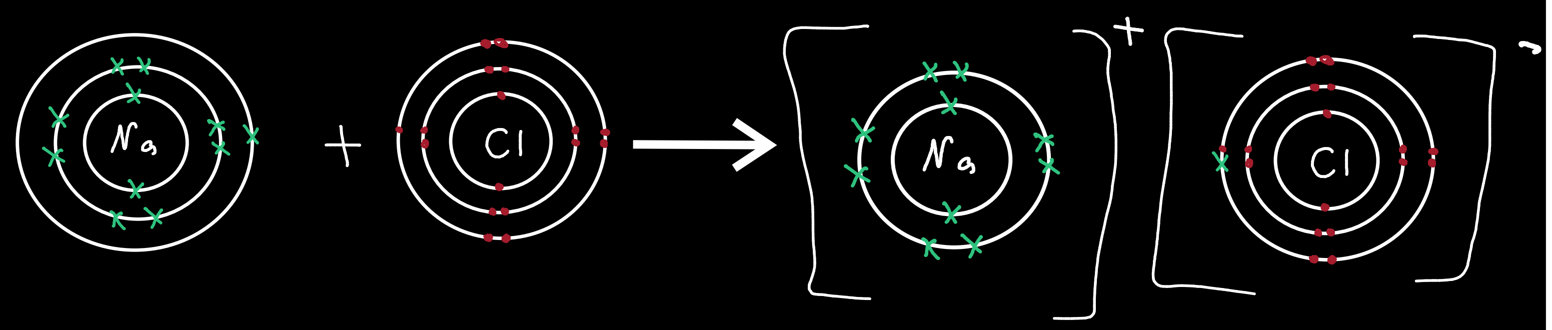
Each ion is trying to form a full other shell (octet rule) and this is achieved by:

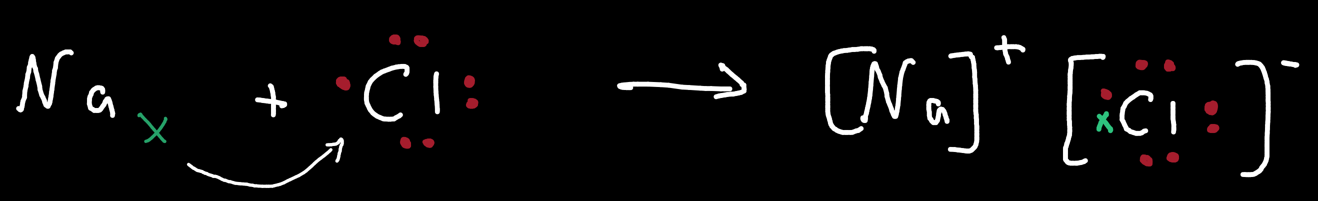
Metals : losing an electron forming postive ions (cations)

Non-metals: gaing an electron forming a negative ion (anion)

Each ions is isoelectronic to the nearest nobel gas

Isoelcetronic: Particles (atoms, ions, or molecules) that have the same number of electrons and the same electron configuration.





Evidence of existance of ions

−Physical properties

1. High Melting and Boiling Points

Explanation: Ions are held by strong electrostatic forces in a giant lattice. A large amount of energy is required to overcome these forces.

Result: This results in a high melting point.

2. Brittleness

General Rule: Ionic compounds are considered to be brittle.

Mechanism: When a force is applied, it can cause layers of ions to shift. This movement places ions of the same charge next to each other, which repel (like charges repel), causing the material to shatter.

3. Electrical Conductivity

Solid State: Not conductive.

−Reason: Ions are fixed in place and there are no delocalised electrons.

Molten or Aqueous Solution: Conductive.

−Reason: The ions are free to move.

Electrolysis: If a current is applied, the compound will undergo electrolysis

Cations → negative electrode (cathode) and are reduced.

Anions → positive electrode (anode) and are oxidized.

4. Solubility

General Rule: Ionic compounds are soluble in water.

Mechanism: The bond breaking is caused by interaction (attraction) between ions and water molecules followed by hydration.

Hydration: The process where water molecules are attracted to ions and surround them due to water's polarity(will be explaind in futher details in covelant bonds)

−Oxygen end (slightly negative) attracted to positive ion.

−Hydrogen end (slightly positive) attracted to negative ion.

−This forms an ion-dipole bond.

−Migration of ions in electrolysis

Like whats mentioned earlier ionic compounds undergo electrolysis when dircet current is passed through it (in its molten or aqueous states)

During this process:

−Positive ions (e.g Na^+) migrate towards the negative electrode where they gain electrons forming atom (e.g Na atoms).

−Negative ions (e.g Cl^-) migrate towards the positive electrode where they lose electrons forming atoms/molecules (e.g Cl atoms).

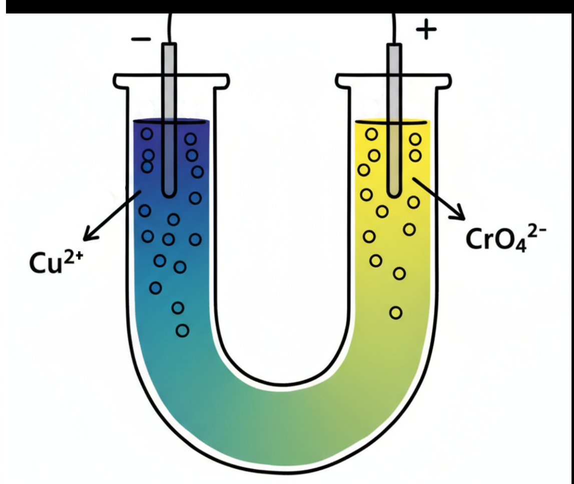
This concept could be demonstrated even futher using colored ions such as :

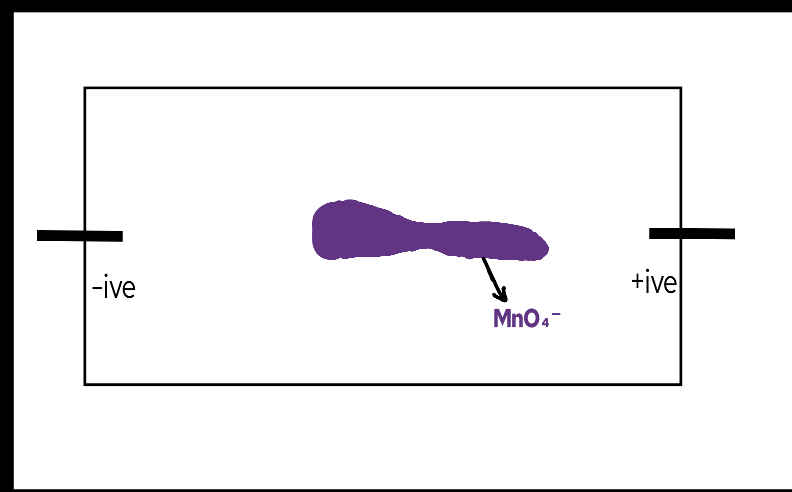
−Copper ion (Cu^{2+}) which has a blue color

−Chromate ion (CrO_4^{2-}) which has a yellow color

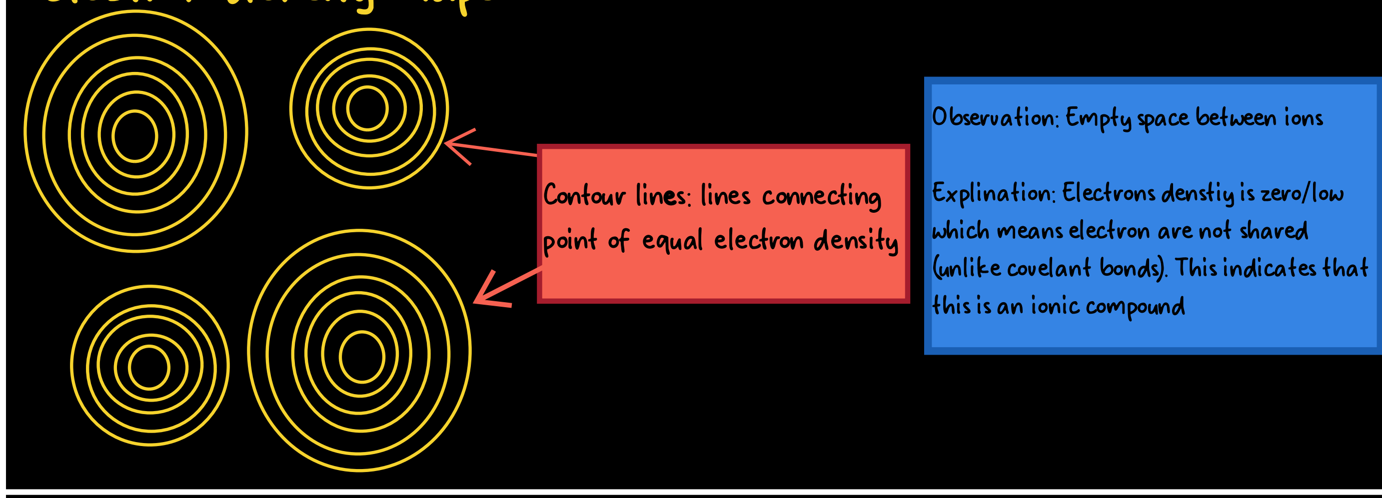
−Manganate ion (MnO_4^-) which has a purple

(During electrolysis, the area around each electrode will turn the color of the ion migrating towards it)





−Electron density maps



Strength of ionic bonding

Lattice Enthalpy(energy): The enthalpy change when 1 mole of ionic compound is fromed for its gaseous ions in standard conditions. (exothermic)

A more Negative enthalpy → Stronger ionic bonding → More energy required to break the bonding

The strength of an ionc bond relies on:

−Ionic raduis

Decreasing the ionic radius of the ions in a compound reduces the distance between them which strengthens (more exothermic) the forces of attraction.(factors affecting the ionic radius where explained in more depth in topic 2 notes)

Example: LiF has a higher lattice enthalpy than KF.

Note: for isoelectronic ions the lower the nuclear charge (number of protons) the greater ionic radius as they has the same electron config and shielding

$N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

−Ionic charge

Increasing the charge of ions in the compound strengthens (more exothermic) the forces of attraction.

Example: MgO (Mg^{2+} & O^{2-}) has a much higher lattice enthalpy than NaCl (Na^+ & Cl^-).

Ionic bonding

Polarisation

Polarisation: the distortion of the electron density (electron cloud) of an anion by a cation so that it's no longer sphereical.

When the lattice enthalpy is calculated theoretically the ions are assumed to be perfectly spherical. However, when it is measured (e.g. using Born–Haber cycle) the energy required is greater(more negative) as the bonds are not 100% ionic (it has some covalent characteristics) caused by the polarixation the cation to the anion

Cation

The polarisating power(ability to distort) of a cation depends on:

–Ionic radius

A samller cation is more polarising than larger one as it makes the outer shell of anion closer to the cations nucleus.

–Charge

The greater the charge of the cation the stronger it attract the outer electrons (more polarising).

The polarisability (ease of polarising) of an anion depends on:

Anion

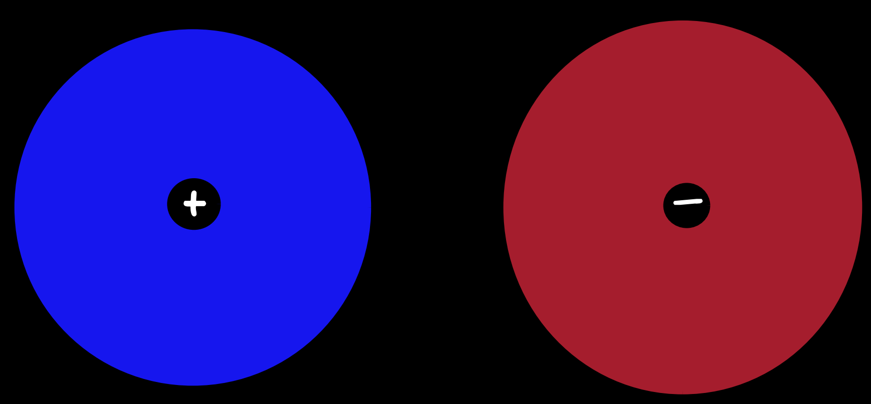
–Ionic radius

A larger anion is more polarisable than smaller one one as the electron cloud would bew futher from its nucleus so easier to distort.

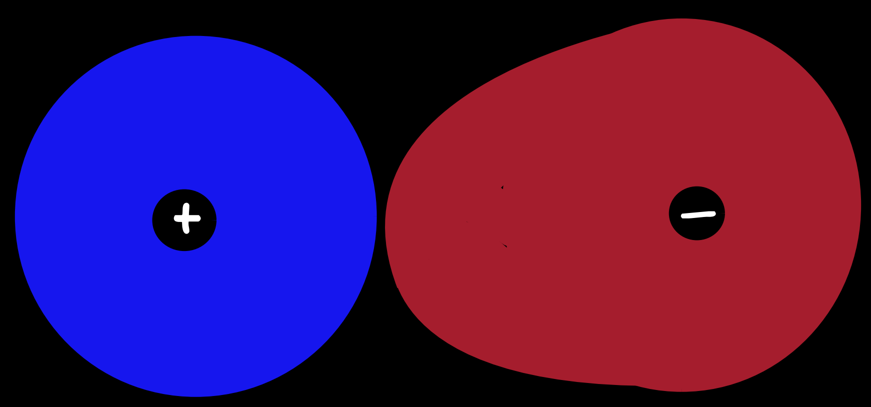
–Charge

The greater the charge of thse anion the more polarisable it would be as there would be more electron to distort.

100% Ionic



Ionic with some covelant characteristics



The polarisation creates some degree of electron sharing (covalent)

Covalent bonding

Covelant bond: the strong electrostatic attraction between the two nuclei of bonded atoms and the shared pair of electrons.

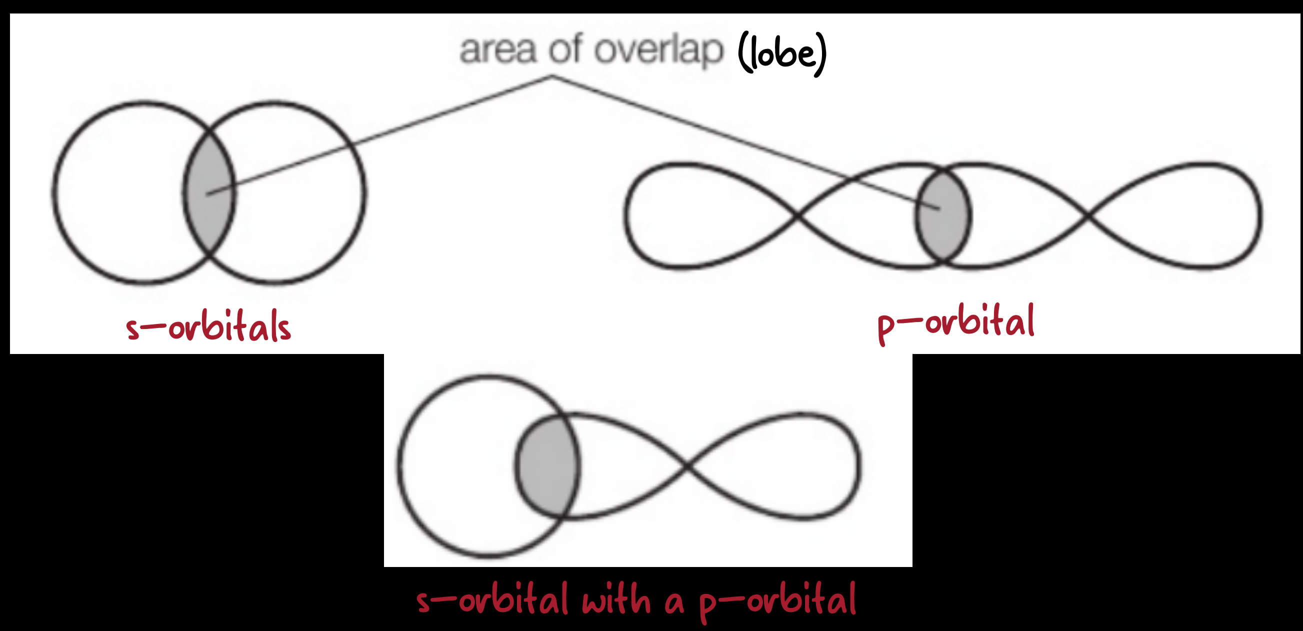
A covalent bond is fromed when two atomic orbitals in the two atoms containing an unpaired (single) electron overlap forming a molecular orbital containing paired (two) electrons.

Types of covalent bonds

–Sigma (σ) bond

A sigma bonds is formed by an end–to–end overlap of atomic orbitals, and the electron concentrated between the nuclei of the bonding atoms along the internuclear axis.

note: the electron density is symmetrical about the line joining nuclei (internuclear).



–Pi (π) bond

A Pi bonds is formed by an side–to–side overlap of atomic orbitals, and the electron concentrated above and below the palne of the nuclei of the bonding atoms.

Pi bonds are only formed after a sigma bond has been formed, so they are only found is doulbe and tirple bonds.

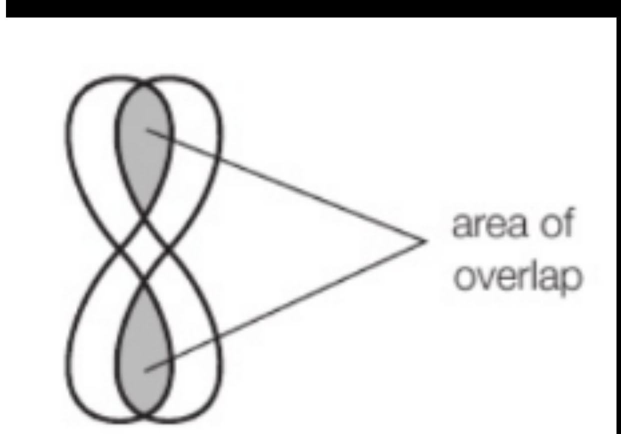
single → 1 sigma

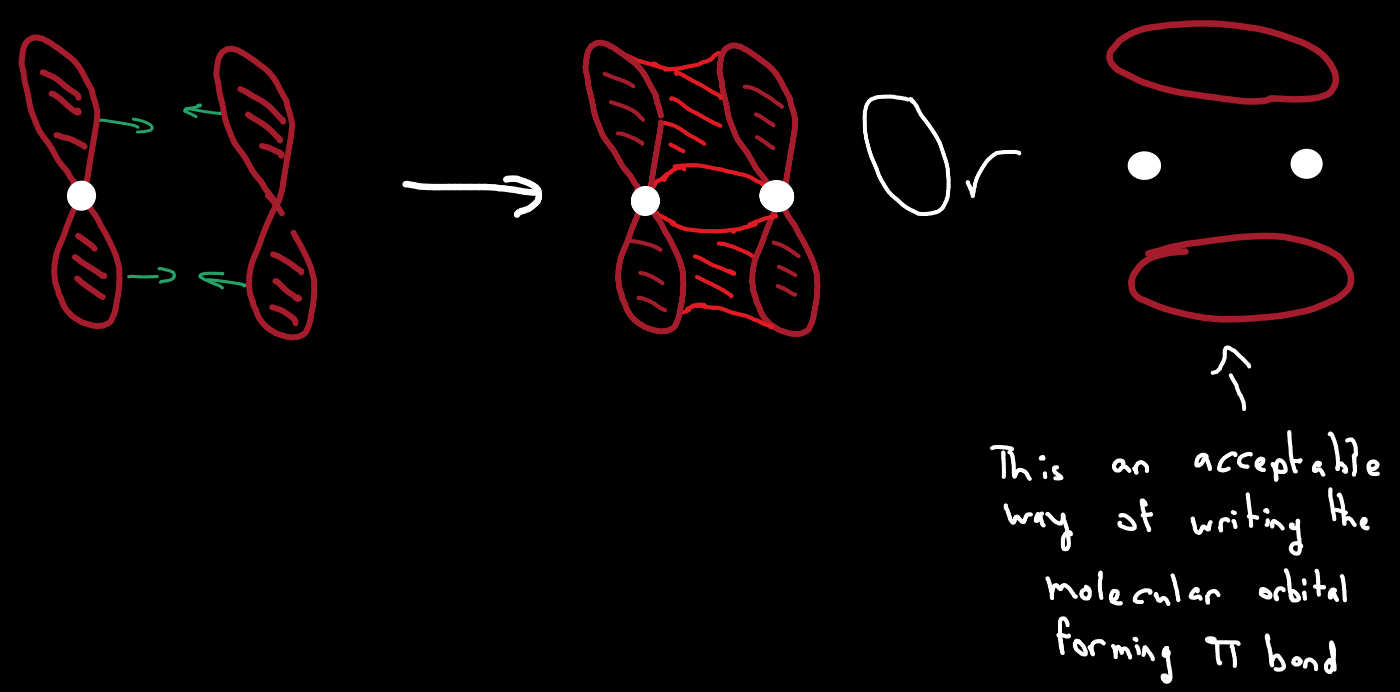
doulbe → 1 sigma + 1 pi

tirple → 1 sigma + 2 pi

Note: a Pi bond is weaker than a Sigma bonds. That is why the reactivity of alkenes > alkanes

Reason: : sigma form an end–to–end overlap which is harder to break compared to side–to–side (Pi).





Covalent bonding

Bond length and strength

Bond length: The distance between two atoms that are covalently bondend together.

The strength of conalent bond relies on the amount of energy required to break 1 mole of the bond is the gaseous state.

—Relation:
Generally the shorter the bond length the stronger the attraction hence bond strength.

BOND	BOND LENGTH / nm	BOND STRENGTH / kJ mol^{-1}
Cl–Cl	0.199	242
Br–Br	0.228	193
I–I	0.267	151
C–C	0.154	347
C=C	0.134	612
C≡C	0.120	838
N–N	0.145	158
N=N	0.120	410
N≡N	0.110	945
O–O	0.148	144
O=O	0.121	498

Fluorine exception:
F–F bond is shorter than Cl–Cl but weaker because:
Small atomic size → The small size forces the lone pairs on the two fluorine atoms extremely close together creating strong repulsion between the them, which weakens the bonding interaction.

Note: when coparing bond strength make sure bond are or similar nature

Electronegativity (E.N.)

Electronegativity: the ability of an atom to attract a bonding pair of electrons in a covalent bond.

most E.N.

									H 2.1								He
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe
Cs 0.7	Ba 0.9	La 1.1	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn

Electronegativity values

observation: non–metals have higher E.N. values.

reason: they have more tendancy to gain electrons

observation: nobel gases have not E.N. values.

reason: They are unreactive (do not loss nor gain electrons) and would rarely form covalent bonds

Factors affecting electronegativity (attraction):
—Proton number
More protons → more attraction
—Shielding by inner electrons
More shielding → less attraction
—Distance from nucleus
Greater distance → less attraction

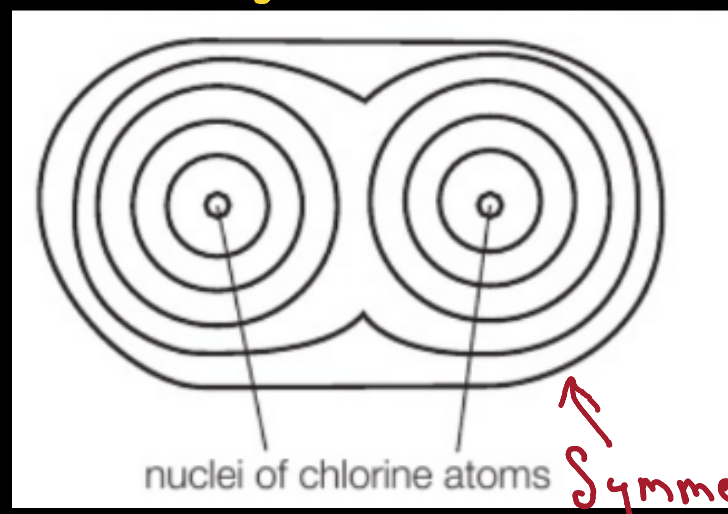
Trends:

—Across the period
More protons + less distance (atomic radius) + same shell (same shielding) → increase in E.N value (more attraction)
—Down the group
Greater distance(atomic radius) + more shells/shielding→ decrease in E.N value (less attraction for bonding electrons)

Non–polar (pure) covalent bonds

If two atoms of same element are bonded the disribution of electrons between the nuclei will be symmetrical this is because they have the same E.N. value so their attraction ability is the same (100% covalent)

—Distribution of electron density

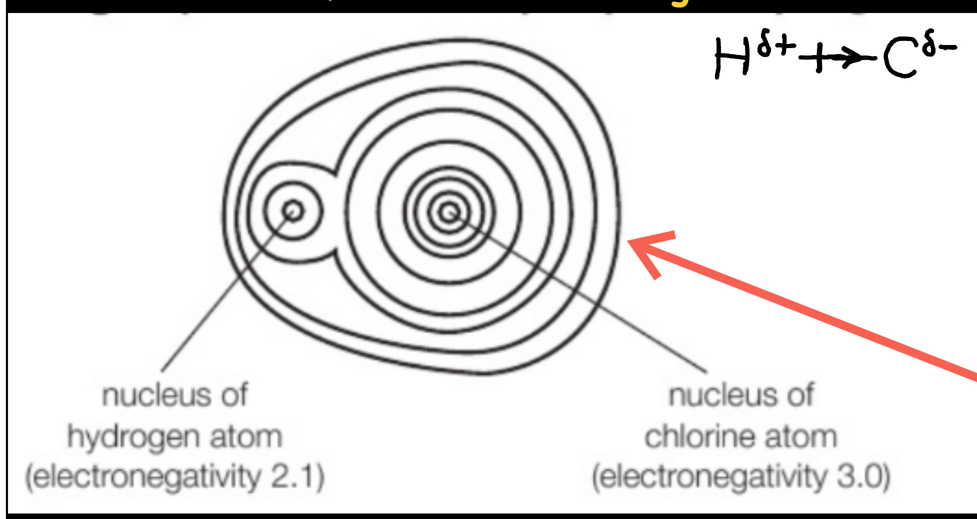


Symmetrical

Polar covalent bonds

If their is a difference isn E.N. values the distribution of electron density won't be symmetrical (unequal distribution).This means there is a separation in charge between two atom so carries a slightly positive(+) charge while other carries a slightly negative(–) charge. (this sepearation is charge in called a dipole)

—Distribution of electron density



δ+ : Slightly positive

δ- : Slightly negative

observation: contour lines are more closely spaced around Cl atom

reason: Cl is more E.N. making electron density around it greater (hence a negative charge)

Contunuum of bonding types

In reality, chemical bonding is not purely ionic or covalent. Instead, there is a spectrum, where bonds have varying degrees of ionic and covalent character. The difference in electronegativity determines the charater degree.

ELECTRONEGATIVITY DIFFERENCE	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1
APPROXIMATE % IONIC CHARACTER	0	0.5	1	2	4	6	9	12	15	19	22	26

ELECTRONEGATIVITY DIFFERENCE	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3
APPROXIMATE % IONIC CHARACTER	30	34	39	43	47	51	55	59	63	67	70	74

ELECTRONEGATIVITY DIFFERENCE	2.4	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2	3.3
APPROXIMATE % IONIC CHARACTER	76	79	82	84	86	88	89	90	91	92

—**Nonpolar** Covalent Bond: Electronegativity difference is approximately 0 to 0.4.
—**Polar** Covalent Bond: Electronegativity difference is greater than 0.4 but less than approximately 1.7–2.0.
—**Ionic** Bond: Electronegativity difference is greater than approximately 1.7–2.0.

Covalent bonding

Drawing (simple) Covalent bonds

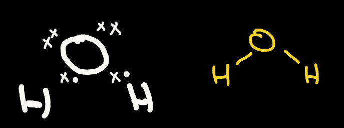
Cross and dot diagram

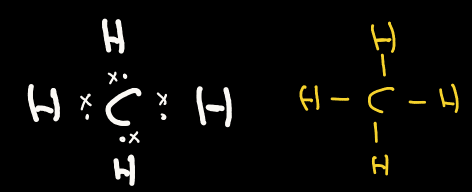
Octet rule: the outer shell electrons in each atoms must be the same as outer shell electrons in a noble gas

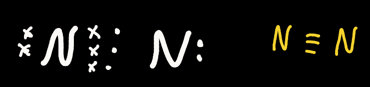
Drawing tips:

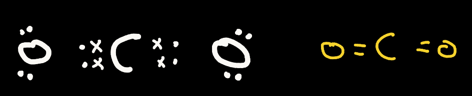
1. The atom with the least electronegativity will be the central atom (exception: hydrogen)
2. Make sure that the total number of electrons stays the same

Examples:

H_2O : 

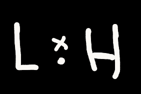
CH_4 : 

N_2 : 

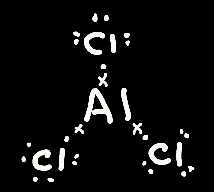
CO_2 : 

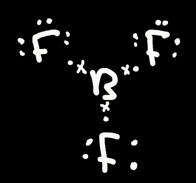
Exceptions to octet rule

Octet Deficient:

LiH : 

$BeCl_2$: 

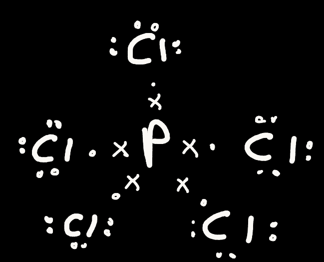
$AlCl_3$: 

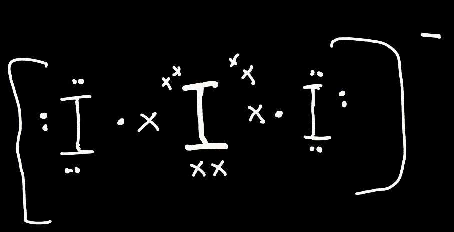
BF_3 : 

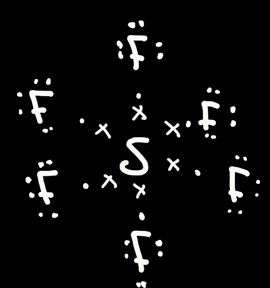
Expanded octet:

Note: an expanded octet only form for element above period 2 as they would have a "d" orpital.

AsF_5 : 

PCl_5 : 

I_3^- : 

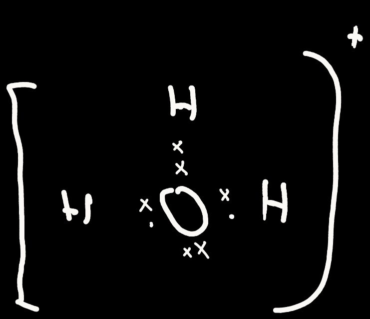
SF_6 : 

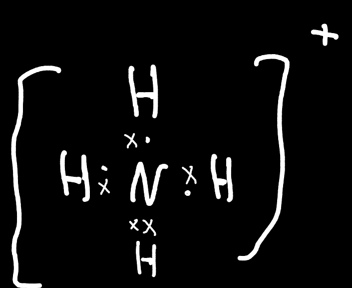
Dative molecules

Dative covalent bond: a pair of electrons shared between two atoms where both of these electrons come from the same atom.

Example

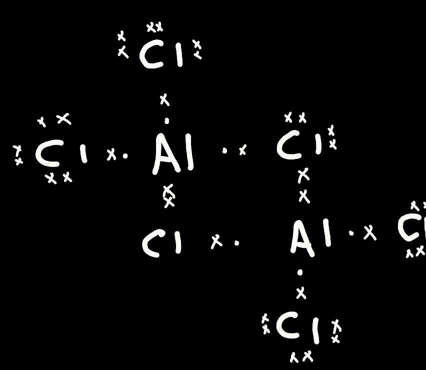
CO : 

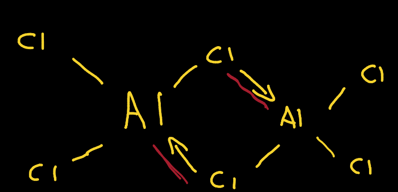
H_3O^+ : 

NH_4^+ : 

Dimer: two identical molecules joined/held by dative bonds

Al_2Cl_6





Shapes of simple molecules and ions

USEPR (valence shell electron pair repulsion)/ERP theory

This theory states:

The shape of a molecule or ion is caused by repulsion between the pairs of electrons, both bonding and lone(non-bonding) pairs that surround the central atom

The electron pairs arrange themselves around the central atom so that repulsions between them is at its minimum.

Lone pair–lone pair repulsion > lone pair–bonding pair repulsion > bond pair–bond pair repulsion

How to explain the shape of the molecule/ion?

1–State number of sets bonding pairs (double bonds are considered one) and lone pairs

2–"They arrange to maximum seperation and minimum repulsion"

3–

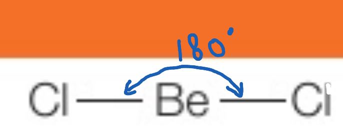
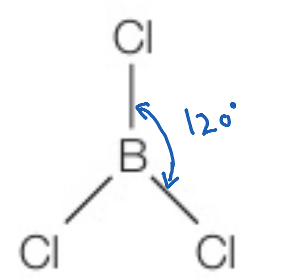
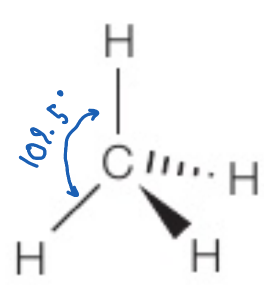
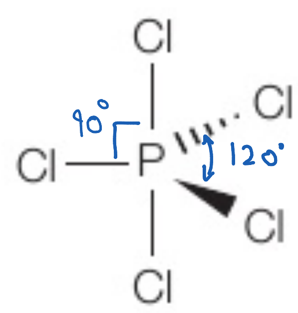
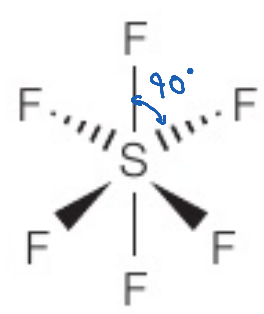
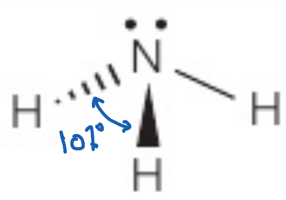
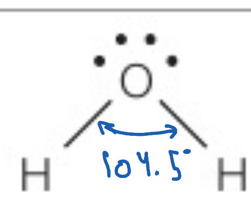
if there are only bonding pairs →"pairs repel equaly"

if there is only 1 lone pairs and rest are bonding →"lone pair–bonding pair repulsion > bond pair–bond pair repulsion"

if there are more than 1 lone pairs and rest are bonding →"lone pair–lone pair repulsion > loner pair pair–bond pair repulsion"

4–State the angle and shape

"Tetrahedral bond angel(109.5) is reduced" in case of V–shaped and pyramidal

NUMBER OF BOND PAIRS	NUMBER OF LONE PAIRS	SHAPE	EXAMPLE
2	0	linear	
3	0	trigonal planar	
4	0	tetrahedral	
5	0	trigonal bipyramidal	
6	0	octahedral	
3	1	trigonal pyramidal	
2	2	V-shaped	

Covalent bonding

Polar and non-polar molecules

Polar Molecules

Electron Density: Uneven (non-symmetrical). Electrons are pulled closer to the more electronegative atoms, creating a "lopsided" molecule.
Polar Bonds: Must have polar bonds (e.g. O-H, N-H, C-F).
Types of bonds: All bonds can be the same type OR they can have different types (which often contributes to the polarity such as $CHCl_3$)
Dipole Cancellation: The dipoles from these bonds DO NOT cancel out due to a non-symmetric shape.
Net Dipole Moment: YES. The molecule has a positive end and a negative end.
Example: Water (H_2O) – The bent shape means the two O-H bond dipoles reinforce each other instead of canceling

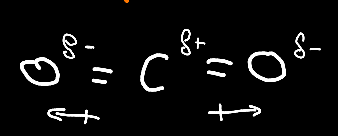
Non-Polar Molecules

Electron Density: Even (symmetrical). The charge is balanced across the molecule.
Polar Bonds: Can have no polar bonds (e.g. Cl_2, H_2) OR can have polar bonds.
Types of bonds: If polar bonds are present, they are typically the same type and arranged symmetrically. (It is very rare to have different bond types and still be non-polar)
Dipole Cancellation: If polar bonds are present, their dipoles DO cancel out due to a highly symmetric shape.
Net Dipole Moment: NO. The molecule has no distinct charged ends.
Example: Carbon Dioxide (CO_2) – The linear shape causes the two polar C=O bond dipoles to point in opposite directions and cancel each other out.

Symmetrical and non symmetrical shapes

Symmetrical:
1.Linear
2.Trigonal planar
3.Tetrahedral
4.Trigonal bipyramidal
5.Octahedral

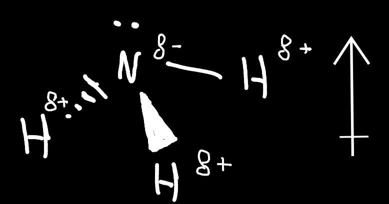
Example



How to explain molecules polarity:
1-State the the shape and whether it is symmetrical or not
2-The type of bonds whetherthey are identical/same or not
3-Determine Dipole Cancellation
4-State net dipole moment

Non-symmetrical:
1.Trigonal pyramidal
2.V-shaped

Example



Molecular structures

Properties

They have low melting and boiling points because the individual molecules are held (within a crystal lattice in solid state) by only weak intermolecular forces, not by strong covalent or ionic bonds. Relatively little energy is needed to overcome these weak forces and break down the orderly lattice structure, allowing the molecules to escape into the liquid or gas phase.

Note: The strength of these intermolecular forces increases with the size of the electron cloud (more electrons) and the length of the molecule.

Giant Covalent structures

General properties

High Melting and Boiling Points

Explanation: Giant covalent structures have very high melting and boiling points because a large amount of energy is required to break the strong, extensive(many) network of covalent bonds throughout the giant covalent lattice structure
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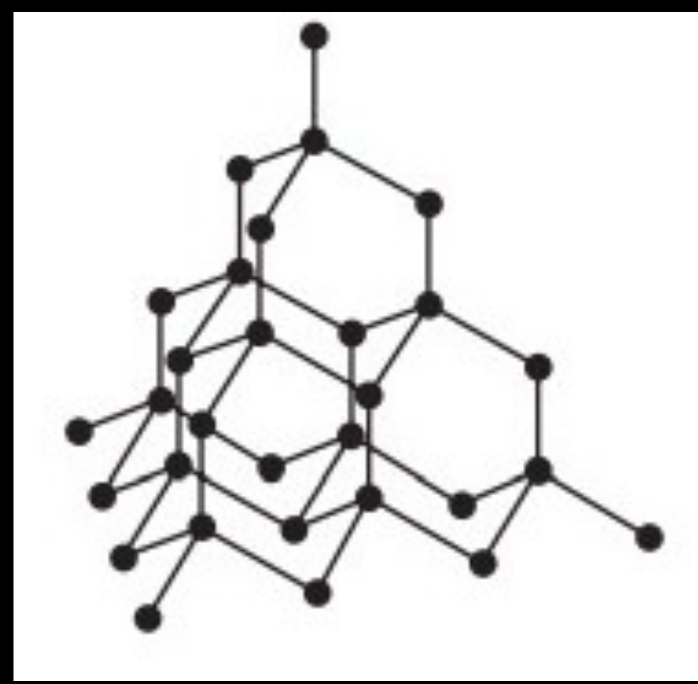
Not Conductive (except graphite/graphene)

Explanation: The outre electrons are fixed in place so there wouldn't be any mobile nor delocalized electrons

Diamond



Description: Each carbon atom is bonded to 4 other carbon atoms (sigma bonds) forming a 3D tetrahedral.
note: all of its bonds are strong covalent bonds with not intermolecular forces making it rigid and strong.

Properties and uses
1-One of the hardest substances known (use: drilling and cutting)
2-It is a poor conductor of electricity (all electrons are shared hence no mobile delocalized electrons)
4-Color from Trace Impurities (use: creates rare and valuable colored gemstones for jewelry)
3-High Optical Dispersion ("Fire") (use: creates the sparkle in colorless gemstones for jewelry)

Silica/ Silicon dioxide (SiO_2)

Description: Is very similar to diamond where each silicon atom is bonded to 4 oxygens atoms atoms (sigma bonds) and each oxygen in bonded to 2 silicon atoms forming a 3D tetrahedral structure around Si.
note: SiO_2 represent the ratio of Si:O not the bonding to it must not be mistaken for simple compounds like CO_2

Silicon dioxide vs Diamond

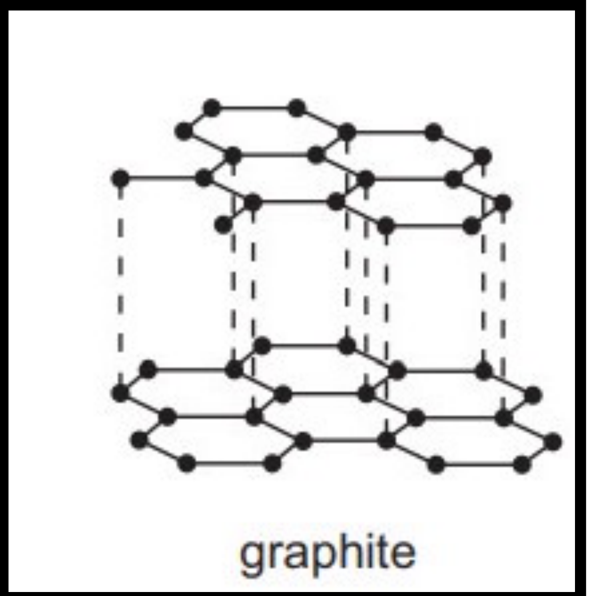
Diamond is a higher melting/ boiling point as there are greater bonds per atom + oxygen(in SiO_2) has lone pairs

Giant Covalent structures

Graphite

Description: Each carbon atom is bonded to 3 other carbon atoms (sigma bonds) forming a layers of hexagonal rings. the layers of hexgonal rings are held by weak intermolecular forces so the layers can slide over each other when a force is applied (soft and malleable)

note: the fourth electron that is not shared becomes delocalized and free to move



Properties and uses

1-Layers can slide over one another/soft dues to weak forces between graphite layers (use: lubricant)

2-Conduct electricity due to mobile delocalized electrons (use: electrodes)

note: unlike metal graphite only conducts in primarily within the planes. The delocalized electrons can move freely across the 2D plane of the hexagonal sheets, but they cannot easily jump between the sheets due to the large separation and weak forces.

3-High melting point due to strong covalent forces withen each layer

Graphene

Description: it is a single layer of graphite where each carbon atom is bonded to 3 other carbon atoms (sigma bonds)

note: the fourth electron that is not shared becomes delocalized and free to move.

Properties and uses

1-Electrical Conductivity and transparent (use: transparent electrodes scores)

2-Uncreactive (use: oxidation resistant layer)

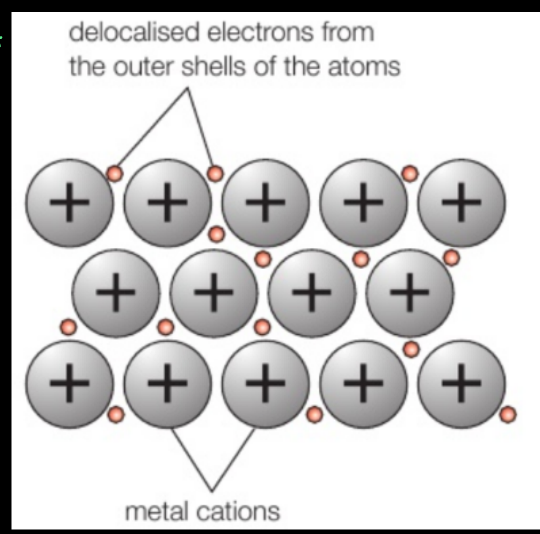
3-Transparent & Conductive (use: flexible electronic such as Touchscreens, Solar Cells)

4- Ultra-Low Friction / Superlubricity (use: low friction coating)

Metallic bonding

Metallic bond: the strong electrostatic attraction between the positive metal ions and the sea delocalized electrons surrounding them.

if asked about a specific ion describe the:
1)Charge of cation
2)Num. of electrons per ion



Description: Metals have characteristic properties due to their giant metallic lattice structure, which consists of a regular arrangement of positive ions (cations) surrounded by a 'sea' of delocalized electrons electrostatically attracted.

Properties

1. High Melting and Boiling Points

Explanation: large amount of energy is required to overcome the strong electrostatic forces of attraction (metallic bonds) between the positive metal ions and the sea of delocalized electrons.

2. Good Conductors of Electricity

Explanation: The delocalized electrons are free to move(mobile) throughout the metallic lattice. When a voltage (potential difference) is applied across a metal, these mobile electrons flow/move forming an electric current.

3. Good Conductors of Heat

Explanation: Heat is transferred efficiently through metals via two main mechanisms:

Electron Movement. The mobile delocalized electrons gain kinetic energy where the metal is heated. They move rapidly throughout the lattice, colliding with ions and other electrons, thereby transferring kinetic (thermal) energy.

Vibrating Ions The positive ions themseleves vibrate in fixed positions. As they vibrate more vigorously, the cations transfer energy to their neighbors through collisions.

4. Malleable and Ductile

Malleable: Can be hammered or rolled into sheets.

Ductile: Can be drawn into wires.

Explanation: The layers of positive ions in the metallic lattice can slide over one another without shattering the structure. The delocalized electrons act as a "glue"(hold togther) that maintains the metallic bonding by moving to accommodate the shift. This prevents layers of like-charged ions from coming directly opposite each other and repelling, allowing the metal to be reshaped permanently.

Strength

Factors affecting the strength of metallic bonding (attraction between cation and delocalized electrons):

1-Number of delocalized electrons (more electrons → stronger bond)

2-Charge (greater charge → stronger bond)

3-Ionic radius (smaller ions → stronger bond)

	GIANT METALLIC	GIANT IONIC	GIANT COVALENT	MOLECULAR
Particles present	positive ions and delocalised electrons	positive and negative ions	atoms	molecules
Type of bonding	metallic	ionic	covalent	covalent
Are there any intermolecular forces of attraction?	no	no	no	yes
Melting and boiling temperatures	fairly high to high	fairly high to high	high to very high	generally low
Electrical conductivity	good when solid and when molten	non-conductor when solid; good when molten	non-conductor	non-conductor
Solubility in water	insoluble unless the metal reacts with water, e.g. sodium	generally soluble, but with notable exceptions, e.g. AgCl, AgBr, AgI and $BaSO_4$	insoluble	generally insoluble, but may dissolve if hydrogen bonding is possible (e.g. sucrose), or if the substance reacts with water (e.g. Cl_2) (See **Topic 7** for an explanation of hydrogen bonding.)